

CMPSC 463 (Spring 2026)

Design Analysis Algorithms

Project 2

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Project Description

The project aims at developing the Humanitarian Routing System, which serves as a means for simulating and optimizing the distribution of aid through the world's transport network. In order to achieve that goal, various types of facilities like warehouses, camps, stores, and hospitals, can be considered as nodes on a graph, and their interconnections will correspond to the possible routes that one might travel. The primary aim of the project lies in determining the safest and most effective route between two chosen facilities. The Humanitarian Routing System considers not only the distance between facilities but also the existence of risk zones, which can disrupt travel.

Significance

The significance of this project is its practical applicability in solving real-world humanitarian logistics issues. In cases of armed conflict, the simple choice of the shortest path might not be sufficient, as some regions may be too risky for navigation. This application offers an innovative way of making routing decisions by giving priority to safe paths, rather than those offering minimal distance only. The significance of the project lies also in its combination of graph theory algorithms and geospatial visualizations in one application. Users can observe how the selected paths depend on particular conditions and how their routing decision-making changes in accordance with these conditions. The risk visualization through heatmaps and the introduction of route analytics like the risk level and routing efficiency offer additional information and make the decision-making process more comprehensive.

Code Structure

The architecture of the project is divided into a modular and structured approach where the backend and frontend are clearly separated from each other. The backend is coded in Python and uses Flask. There are two modules in the backend part of the project – the first module codes the creation of the graph and routing algorithm while the second one implements the APIs for getting the required data and calculating routes. The frontend is written using HTML and JavaScript where the HTML page contains the code of the user interface and layout while the JavaScript page performs all calculations related to rendering the map, user interaction, calculating the route, and analyzing its properties.

Algorithms Description

The main algorithm employed in this project is the Dijkstra's shortest path algorithm. Dijkstra's algorithm, in its standard form, seeks to find the optimal path connecting two points with minimum overall distance. In this case, the algorithm has been adapted to take into consideration risk factors and applies a penalty whenever the routes intersect or are near risky zones. The overall cost of the route will thus be the summation of the distances between the nodes and any extra penalty that comes from risky zones. The Haversine algorithm has been incorporated into the program to calculate geographical distances.

Verification

For checking the validity of the algorithm, various basic test cases have been carried out. For example, in a smaller graph having only a few numbers of nodes, with the distance among nodes being known, the modified algorithm was able to successfully calculate the shortest path, where there is no risk involved. However, in case a risk factor is considered and a risk zone is

introduced in the shortest path calculated by the modified algorithm, then the modified algorithm selects another route, which has a lower cost considering all possible penalties.

Results and Analysis

The system delivers both graphical and numerical outcomes that signify its capability to work according to design purposes. On the map interface, nodes are presented using different icons according to their categories (warehouse, camp, store, and hospital). Areas under the zones of risks are also illustrated in the system with the help of the heatmap, thus making it easy for one to spot regions of high risks. As soon as two nodes are selected by the user, the system will calculate the route connecting the two nodes through highlighted paths and moving trucks along the path on the map interface. Apart from the graphical information, the system provides information related to analytics including distance covered, number of stops, time taken, efficiency, number of risky zones passed through, and total amount of risk.

Conclusion and Limitation

Conclusively, it is evident from this project that classical graph algorithms can be modified to solve practical problems associated with risk and uncertainty. The modification of the Dijkstra's algorithm to account for risk by incorporating the dangers into the calculation has enabled this algorithm to produce safer and more practical routes. Moreover, through visualization tools, this project enables one to see and understand better what the algorithm does and what kind of information it generates. Some of the challenges faced in this project include the use of artificially generated data sets and not accounting for other constraints like time windows and vehicle capacities. Also, while generating nodes randomly, some nodes ended up being located outside the intended map boundaries and this made it impossible to have realistic and

visualizable networks. In this regard, when the nodes' locations were randomly generated without any restrictions, some nodes ended up being in isolated places on the map because of the great variability in their coordinates. Consequently, the routes would be unnecessarily long and hard to visualize. Therefore, this problem was solved through generating nodes within specified geographic areas. Despite these limitations, the project effectively applies concepts from graph theory, optimization, and data visualization, showing how they can be combined in a practical application.

AI Usage

The application of AI software such as ChatGPT and Claude to help develop this project was only meant to serve as a supplementary measure. The AI software was instrumental in helping out with the debugging, and user interface design. The AI software also recommended several ways that the routing algorithm can be improved upon.

Overall

The project is an indicator of my knowledge of graph algorithms and their use in practical situations. Algorithm development combined with visualization creates an efficient yet easy to use tool. The project is relevant to course material, especially to such themes as algorithms for solving shortest paths problems and optimizations techniques. In addition to technical issues, the report also describes the development process, design choices, and implementation details.